

Entrega T10

Métodos Matemáticos Para la Física

1. Evaluar las siguientes integrales sobre la región $R = [0, 1] \times [0, 1]$.

a) $\int_R y e^{xy} dA$

b) $\int_R (ax + by + c) dx dy$

$$\begin{aligned} a) \int_R y e^{xy} dx dy &= \int_0^1 \int_0^1 y e^{xy} dx dy = \int_0^1 [e^{xy}]_0^1 dy = \int_0^1 e^y - 1 dy \\ &= [e^y - y]_0^1 = e - 1 - 1 = e - 2 \end{aligned}$$

$$b) \int_0^1 \int_0^1 (ax + by + c) dy dx = \int_0^1 [axy + by^2/2 + cy]_0^1 dx = \int_0^1 (ax + b/2 + c) dx$$

$$\left[\frac{ax^2}{2} + \frac{b}{2}x + cx \right]_0^1 = \frac{a+b}{2} + c$$

2. Calcular el volumen del solido acotado por la gráfica $z = x^2 + y$, y el rectángulo $R = [0, 1] \times [1, 2]$ y los lados verticales de R

$$\begin{aligned} \int_0^1 \int_1^2 x^2 + y dy dx &= \int_0^1 [x^2y + y^2/2]_1^2 dx = \int_0^1 (2x^2 + 2 - x^2 - 1/2) dx = \int_0^1 (x^2 + 3/2) dx \\ &= \left[\frac{x^3}{3} + \frac{3}{2}x \right]_0^1 = \frac{11}{6} \end{aligned}$$

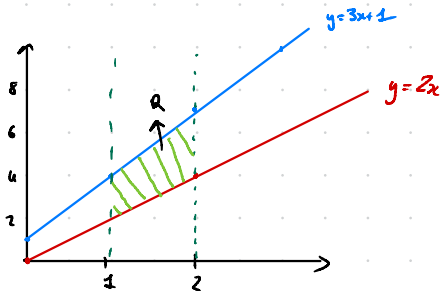
3. Evaluar las siguientes integrales iteradas y trazar las regiones D determinadas por los límites.

a) $\int_1^2 \int_{2x}^{3x+1} dy dx$

b) $\int_0^1 \int_{y^2}^y (x^n + y^m) dx dy$ $m, n > 0$

c) $\int_{-1}^1 \int_{-2|x|}^{|x|} e^{x+y} dy dx$

a) $\int_1^2 \int_{2x}^{3x+1} dy dx = \int_1^2 [y]_{2x}^{3x+1} dx = \int_1^2 (x+1) dx = \left[\frac{x^2}{2} + x \right]_1^2 = \frac{5}{2}$

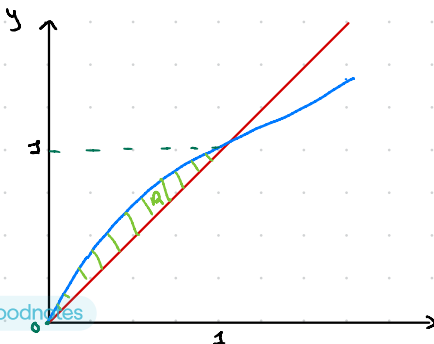


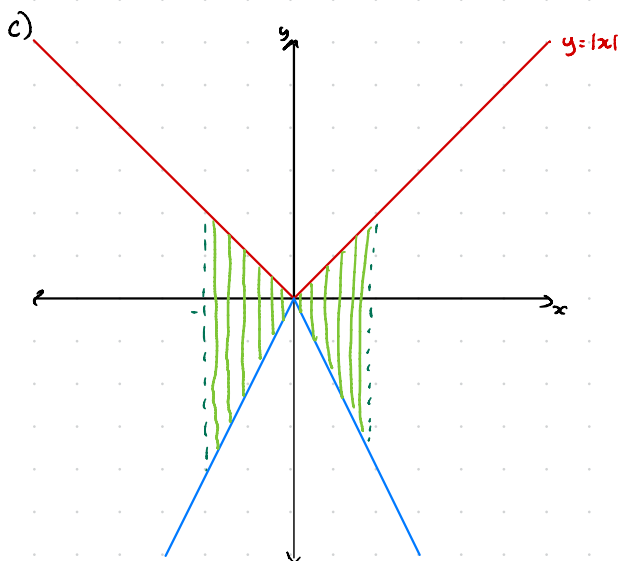
b) $\int_0^1 \int_{y^2}^y (x^n + y^m) dx dy = \int_0^1 \left[\frac{1}{n+1} x^{n+1} + y^m x \right]_{y^2}^y dy =$

$= \int_0^1 \left(\frac{1}{n+1} y^{n+1} + y^{m+1} - \frac{1}{n+1} y^{2(n+1)} - y^{m+2} \right) dy$

$= \left[\frac{1}{(n+1)(n+2)} y^{n+2} + \frac{1}{(m+2)} y^{m+2} - \frac{1}{(n+1)(2n+3)} y^{2n+3} - \frac{1}{m+3} y^{m+3} \right]_0^1$

$= \frac{1}{(n+1)(n+2)} + \frac{1}{m+2} - \frac{1}{(n+1)(2n+3)} - \frac{1}{m+3}$





$$\int_{-1}^1 \int_{-2|x|}^{|x|} e^{x+y} dy dx = \int_{-1}^0 \int_{2x}^{-x} e^{x+y} dy dx + \int_0^1 \int_{-2x}^x e^{x+y} dy dx = 2 \int_0^1 \int_{-2x}^x e^{x+y} dy dx$$

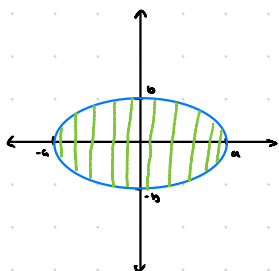
Por simetria

$$= 2 \int_0^1 [e^{x+y}]_{-2x}^x dx = 2 \int_0^1 e^{2x} - e^{-x} dx = 2 \left[\frac{e^{2x}}{2} + e^{-x} \right]_0^1 = 2 \left[\frac{e^2}{2} + \frac{1}{e} - \frac{1}{2} - 1 \right]$$

$$= e^2 + \frac{2}{e} - 3$$

4. Usar integrales dobles para calcular el área de una elipse con semiejes de longitud a y b .

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \rightarrow \frac{y^2}{b^2} = 1 - \frac{x^2}{a^2} \rightarrow y^2 = b^2 - \frac{x^2}{a^2} \rightarrow y = \pm \sqrt{b^2 - \frac{x^2}{a^2}}$$



$$\int_{-a}^a \int_{-\sqrt{b^2 - \frac{x^2}{a^2}}}^{\sqrt{b^2 - \frac{x^2}{a^2}}} dy dx$$

Cambio a coordenadas polares para elipses:

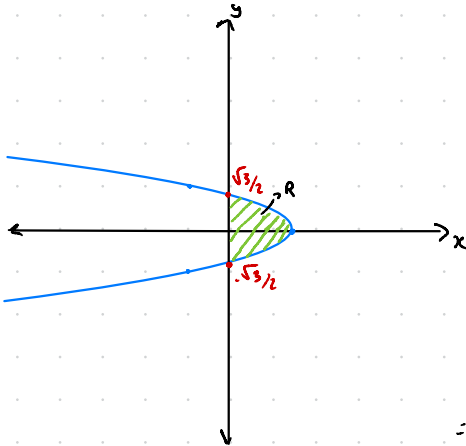
$$\begin{aligned} x &= a \cos \theta & r &\in [0, 1] \\ y &= b \sin \theta & \theta &\in [0, 2\pi] \end{aligned}$$

$$\begin{aligned} \int_0^1 \int_0^{2\pi} ab r dr d\theta &= \int_0^1 ab r dr \cdot \int_0^{2\pi} d\theta = \left[\frac{abr^2}{2} \right]_0^1 [0]_0^{2\pi} \\ &= \frac{ab}{2} \cdot 2\pi = ab\pi \end{aligned}$$

$$dy dx = ab r dr d\theta$$

5. Sea D la región acotada por el eje y la parábola $x = -4y^2 + 3$, calcular

$$\int_D x^3 y \, dx \, dy$$



$$x = r \cos \theta \quad r \in [0, 1]$$

$$y = \frac{\sqrt{3}}{2} r \sin \theta \quad \theta \in [0, \pi]$$

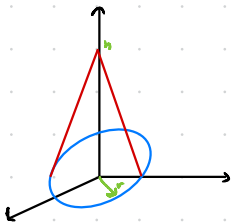
$$J = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \frac{\sqrt{3}}{2} \sin \theta & \frac{\sqrt{3}}{2} r \cos \theta \end{vmatrix} = \frac{\sqrt{3}}{2} r \cos^2 \theta + \frac{\sqrt{3}}{2} r \sin^2 \theta = \frac{\sqrt{3}}{2} r$$

$$\int_0^\pi \int_0^1 r^3 \cos^3 \theta \frac{\sqrt{3}}{2} r \sin \theta \cdot \frac{\sqrt{3}}{2} r \, dr \, d\theta = \int_0^\pi \int_0^1 \frac{3}{4} r^5 \cos^3 \theta \sin \theta \, dr \, d\theta$$

$$= \int_0^\pi \left[\frac{1}{8} r^6 \cos^3 \theta \sin \theta \right]_0^1 d\theta = \int_0^\pi \frac{1}{8} \cos^3 \theta \sin \theta \, d\theta$$

$$= \left[-\frac{1}{32} \cos^4 \theta \right]_0^\pi = \frac{1}{32} - \frac{1}{32} = 0$$

6. Calcular el volumen de un cono de base de radio r y altura h .



Coordenadas cilíndricas (ρ, θ, z)

$$x = \rho \cos \theta \quad \rho \in [0, r]$$

$$y = \rho \sin \theta \quad \theta \in [0, 2\pi]$$

$$z = z \quad z \in [0, h]$$

$$z = \frac{h}{r} \sqrt{x^2 + y^2}$$

$$z = \frac{h}{r} \rho$$

$$\int_0^r \int_0^{2\pi} \int_0^{h \cdot \frac{\rho}{r}} \rho \, dz \, d\theta \, d\rho = \int_0^r \int_0^{2\pi} \int_0^{h \cdot \frac{\rho}{r}} \rho \, dz \, d\theta \, d\rho$$

$$= \int_0^r \int_0^{2\pi} [p z]_0^{\frac{h}{r} p} = \int_0^r \int_0^{2\pi} p h - \frac{h}{r} p^2 d\theta dr$$

$$= \int_0^r [p h - \frac{h}{r} p^2] dr \cdot \int_0^{2\pi} d\theta$$

$$= \left[\frac{p^2 h}{2} - \frac{h p^3}{3r} \right]_0^r \cdot [0]_0^{2\pi} = 2\pi \left(\frac{r^2 h}{2} - \frac{r^3 h}{3r} \right) = \pi r^2 h - \frac{2\pi r^3 h}{3} = \frac{3\pi r^3 h}{3r} - \frac{2\pi r^3 h}{3r} = \frac{\pi r^2 h}{3}$$

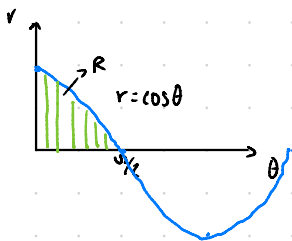
7. En las integrales siguientes, cambiar el orden de integración, esbozar las regiones correspondientes y evaluar las integrales de las dos maneras

a) $\int_0^{\pi/2} \int_0^{\cos \theta} \cos \theta dr d\theta$

b) $\int_0^4 \int_{y/2}^2 e^{x^2} dx dy$

a) $\int_0^{\pi/2} \int_0^{\cos \theta} \cos \theta dr d\theta = \int_0^{\pi/2} [\cos \theta r]_0^{\cos \theta} d\theta = \int_0^{\pi/2} \cos^2 \theta d\theta = \frac{1}{2} \int_0^{\pi/2} \cos 2\theta + 1 d\theta$

$$\frac{1}{2} \left[\frac{1}{2} \sin(2\theta) + \theta \right]_0^{\pi/2} = \frac{\pi}{4}$$



$$\int_0^1 \int_0^{\arccos(r)} \cos \theta d\theta dr = \int_0^1 [\sin(\theta)]_0^{\arccos(r)} dr = \int_0^1 \sin(\arccos(r)) dr$$

$$t = \arccos(r) \rightarrow r = \cos(t) \rightarrow \sin^2(t) + \cos^2(t) = 1$$

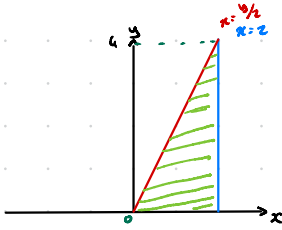
$$\sin^2(t) + r^2 = 1 \rightarrow \sin(t) = \sqrt{1-r^2}$$

$$\int_0^1 (1-r^2)^{1/2} dr = \int_0^1 \sqrt{1-\sin^2(t)} \cdot \cos(t) dt = \int_0^1 \cos^2(t) dt = \frac{1}{2} \int_0^1 \cos(2t) + 1 dt$$

$$r = \cos(t) \quad dr = -\sin(t) dt$$

$$= \left[\frac{\sin(t)}{4} + \frac{t}{2} \right] = \left[\frac{\sin(\arccos(r))}{4} + \frac{\arccos(r)}{2} \right]_0^1 = \frac{\pi}{4}$$

$$b) \int_0^4 \int_{y/2}^2 e^{x^2} dx dy \rightarrow \text{No } \int e^{x^2} dx$$



$$\begin{aligned} \int_0^2 \int_0^{2x} e^{x^2} dy dx &= \int_0^2 [y e^{x^2}]_0^{2x} dx = \int_0^2 2x e^{x^2} dx \\ &= [e^{x^2}]_0^2 = e^4 - 1 \end{aligned}$$

8. Si $f(x, y) = e^{\sin(x+y)}$ y $D = [-\pi, \pi] \times [-\pi, \pi]$: demostrar la siguiente desigualdad

$$\frac{1}{e} \leq \frac{1}{4\pi^2} \int_D f(x, y) dA \leq e$$

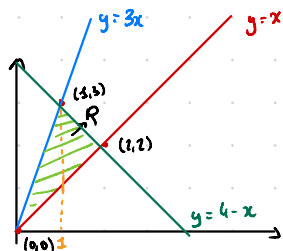
$$\text{Area } D = 2\pi \cdot 2\pi = 4\pi^2$$

$$-1 \leq \sin(x, y) \leq 1 \rightarrow e^{-1} \leq e^{\sin(x, y)} \leq e^1$$

$\int_D f(x, y) = \text{Area de } f(x, y) \text{ sobre la región}$

$$e^{-1} \cdot 4\pi^2 \leq \int_D e^{\sin(x, y)} \leq 4\pi^2 e^1 \rightarrow e^{-1} \leq \frac{1}{4\pi^2} \int_D f(x, y) \leq e^1 \text{ como pide}$$

9. Evaluar $\iint_D e^{x-y} dx dy$, donde D es el interior del triángulo de vértices (0, 0), (1, 3) y (2, 2).



$$\iint_D e^{x-y} = \int_0^1 \int_x^{3x} e^{x-y} dy dx + \int_1^2 \int_x^{4-x} e^{x-y} dy dx$$

$$\int_0^1 \int_x^{3x} e^{x-y} = \int_0^1 [-e^{x-y}]_x^{3x} dx = \int_0^1 1 - e^{2x} dx$$

$$= [x - \frac{1}{2} e^{2x}]_0^1 = \frac{1}{2} - \frac{1}{2e^2} = \frac{1}{2} \left(1 - \frac{1}{e^2}\right)$$

$$\int_1^2 \int_x^{4-x} e^{x-y} dy dx = \int_1^2 [-e^{x-y}]_x^{4-x} dx = \int_1^2 1 - e^{2x-4} dx = [x - \frac{1}{2} e^{2x-4}]_1^2 = \frac{1}{2} \left(1 + \frac{1}{e^2}\right)$$

$$\iint_D e^{x-y} dA = \frac{1}{2} \left(1 - \frac{1}{e^2}\right) + \frac{1}{2} \left(1 + \frac{1}{e^2}\right) = 1 + \frac{1}{e^2}$$

10. Evaluar $\int_W e^{-xy} y dV$, donde $W = [0, 1] \times [0, 1] \times [0, 1]$

$$\int_0^1 \int_0^1 \int_0^1 y e^{-xy} dx dy dz = \int_0^1 \int_0^1 [-e^{-xy}]_0^1 dy dz = \int_0^1 \int_0^1 1 - e^{-y} dy dz = \int_0^1 [y + e^{-y}]_0^1 dz$$

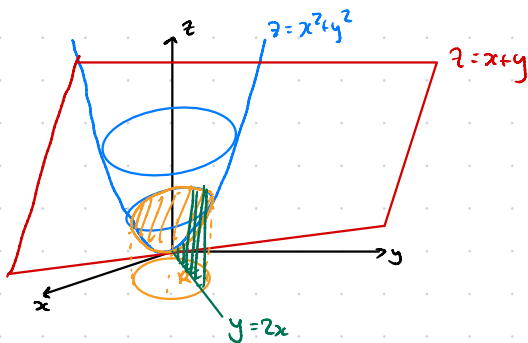
$$= \int_0^1 1 + e^{-1} - 1 dz = \int_0^1 e^{-1} dz = [e^{-1} z]_0^1 = e^{-1}$$

11. Evaluar $\int_W x^2 \cos z \, dV$ donde W es la región acotada por los planos $z = 0$, $z = \pi$, $y = 0$, $y = 1$, $x = 0$ y $x + y = 1$.

$$\begin{aligned} 0 &\leq z \leq \pi \\ 0 &\leq y \leq 1 \\ 0 &\leq x \leq 1-y \end{aligned} \quad \int_0^\pi \int_0^1 \int_0^{1-y} x^2 \cos z \, dx \, dy \, dz$$

$$\begin{aligned} &= \int_0^\pi \int_0^1 \left[\frac{x^3}{3} \cos z \right]_0^{1-y} dy \, dz = \int_0^\pi \int_0^1 \frac{(1-y)^3}{3} \cos z \, dy \, dz = \int_0^\pi \left[-\frac{(1-y)^4}{12} \cos z \right]_0^1 dz \\ &= \int_0^\pi \frac{1}{12} \cos z \, dz = \left[\frac{1}{12} \sin z \right]_0^\pi = 0 \end{aligned}$$

12. Evaluar $\int_0^1 \int_0^{2x} \int_{x^2+y^2}^{x+y} dz \, dy \, dx$ y esbozar la región de integración.



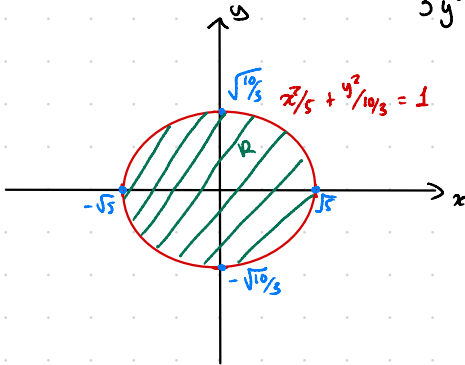
$$\int_0^1 \int_0^{2x} \int_{x^2+y^2}^{x+y} 1 \, dz \, dy \, dx$$

$$\begin{aligned} \int_0^1 \int_0^{2x} \left[z \right]_{x^2+y^2}^{x+y} dy \, dx &= \int_0^1 \int_0^{2x} (x - x^2 + y - y^2) dy \, dx = \int_0^1 \left[xy - x^2y + \frac{y^2}{2} - \frac{y^3}{3} \right]_0^{2x} dx \\ &= \int_0^1 \left(4x^2 - \frac{4}{3}x^3 \right) dx = \left[\frac{4}{3}x^3 - \frac{1}{3}x^4 \right]_0^1 = \frac{1}{6} \end{aligned}$$

13. Hallar el volumen de la región acotada por las superficies $z = x^2 + y^2$ y $z = 10 - x^2 - 2y^2$. Esbozar la región.

$$x^2 + y^2 = 10 - x^2 - 2y^2 \rightarrow 2x^2 + 3y^2 = 10 \rightarrow \frac{x^2}{5} + \frac{y^2}{10/3} = 1$$

$$3y^2 = 10 - 2x^2 \rightarrow y = \pm \sqrt{\frac{10 - 2x^2}{3}}$$



$$\int_{-5}^5 \int_{\sqrt{\frac{10-2x^2}{3}}}^{\sqrt{\frac{10-2x^2}{3}}} dy dx$$

$$x = \sqrt{5} r \cos \theta$$

$$y = \sqrt{\frac{10}{3}} r \sin \theta$$

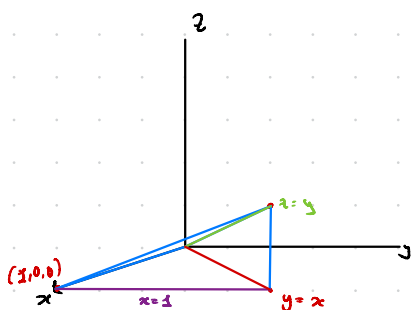
$$J = \left| \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} \right| = \left| \begin{vmatrix} \sqrt{5} \cos \theta & -\sqrt{5} r \sin \theta \\ \sqrt{\frac{10}{3}} \sin \theta & \sqrt{\frac{10}{3}} r \cos \theta \end{vmatrix} \right| = \frac{5\sqrt{6}}{3} r \cos^2 \theta + \frac{5\sqrt{6}}{3} r \sin^2 \theta = \frac{5\sqrt{6}}{3} r$$

$$\iint_D 10 - 2x^2 - 3y^2 dA = \int_0^{2\pi} \int_0^1 (10 - 10 r^2 \cos^2 \theta - 10 r^2 \sin^2 \theta) \frac{5\sqrt{6}}{3} r dr d\theta = \int_0^{2\pi} \int_0^1 (10r - 10r^3) \frac{5\sqrt{6}}{3} dr d\theta$$

$$= \int_0^{2\pi} \left[5r^2 - \frac{5}{2} r^4 \right]_0^1 \cdot \frac{5\sqrt{6}}{3} d\theta = \int_0^{2\pi} \frac{25\sqrt{6}}{6} d\theta = \frac{25\sqrt{6}\pi}{3}$$

14. Cambiar el orden de integración en $\int_0^1 \int_0^x \int_0^y f(x, y, z) dz dy dx$ para obtener otras cinco formas de respuesta. Esbozar la región de integración.

$$\left. \begin{array}{l} 0 \leq x \leq 1 \\ 0 \leq y \leq x \\ 0 \leq z \leq y \end{array} \right\} z \leq y \leq x$$



$$a) \int_0^1 \int_y^1 \int_0^y f(x, y, z) dz dx dy$$

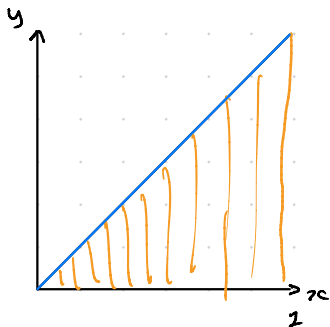
$$b) \int_0^1 \int_0^x \int_z^x f(x, y, z) dy dz dx$$

$$c) \int_0^1 \int_z^1 \int_y^1 f(x, y, z) dx dy dz$$

$$d) \int_0^1 \int_z^1 \int_z^x f(x, y, z) dy dx dz$$

$$e) \int_0^1 \int_z^1 \int_y^1 f(x, y, z) dx dz dy$$

15. Se D la región $0 \leq y \leq x$ y $0 \leq x \leq 1$. Evaluar $\int_D (x+y) dx dy$ haciendo el cambio de variables $x = u+v$, $y = u-v$. Verificar la respuesta obtenida evaluando directamente la integral, usando una integral iterada.



$$\begin{array}{l} x = u+v \\ y = u-v \end{array}$$

$$|J| = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -2$$

$$\int_0^1 (x+y) dx dy = \int_0^1 4u du dv$$

$$0 \leq u-v \leq u+v$$

$$0 \leq u+v \leq 1$$

$$0 \leq u-v \rightarrow v \leq u$$

$$u-v \leq u+v \rightarrow -v \leq v$$

$$0 \leq u+v \leq 1 \rightarrow u+v \leq 1: v \leq 1-u$$

$$v \leq u \leq 1-u \rightarrow u \leq 1/2$$

$$\text{Si } u \leq 1/2 \rightarrow v \leq u \quad (u \leq 1-u)$$

$$\text{Si } u > 1/2 \rightarrow v \leq 1-u \quad (u > 1-u)$$

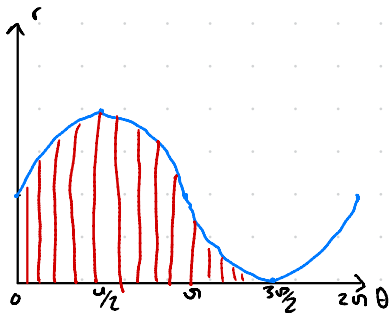
$$\int_D -4uv \, du \, dv = \int_0^{1/2} \int_0^u -4u \, dv \, du + \int_{1/2}^1 \int_0^{1-u} -4u \, dv \, du$$

$$\int_0^{1/2} \int_0^u -4u \, dv \, du = \int_0^{1/2} [-4uv]_0^u = \int_0^{1/2} -4u^2 \, du = \left[-\frac{4}{3}u^3 \right]_0^{1/2} = -\frac{1}{6}$$

$$\int_{1/2}^1 \int_0^{1-u} -4u \, dv \, du = \int_{1/2}^1 [-4uv]_0^{1-u} \, du = \int_{1/2}^1 -4u + 4u^2 \, du = \left[-2u^2 + \frac{4}{3}u^3 \right]_{1/2}^1$$

$$= \frac{2}{3} - \frac{1}{3} = \frac{1}{3} \rightarrow \int_D (x+y) \, dx \, dy = \frac{1}{3} + \frac{1}{6} = \frac{1}{2}$$

16. Usar integrales dobles para hallar el área dentro de la curva $r = 1 + \sin \theta$.



$$\int_0^{2\pi} \int_0^{1+\sin \theta} r \, dr \, d\theta = \int_0^{2\pi} \left[\frac{r^2}{2} \right]_0^{1+\sin \theta} d\theta = \int_0^{2\pi} \frac{(1+\sin \theta)^2}{2} d\theta$$

$$\int_0^{2\pi} \frac{(1+\sin \theta)^2}{2} d\theta \rightarrow \int_0^{2\pi} \frac{1+2\sin \theta + \sin^2 \theta}{2} d\theta = \frac{1}{2} \int_0^{2\pi} 1 + 2\sin \theta + \frac{1-\cos 2\theta}{2} d\theta$$

$$= \frac{1}{2} \left[\theta - 2\cos \theta + \frac{\sin 2\theta}{2} \right]_0^{2\pi} = \left[\frac{\pi}{2} - 2 + 2 \right]_0^{2\pi} = \frac{3\pi}{4}$$

17. Sea D el disco unitario. Expresar $\int_D (1+x^2+y^2)^{3/2} dx dy$ como una integral sobre el rectángulo $[0, 1] \times [0, 2\pi]$ y evaluar.

$$\int_0^{2\pi} \int_0^1 (1+x^2+y^2)^{3/2} dx dy$$

uso coordenadas polares:

$$\begin{aligned} x &= r \cos \theta \\ y &= r \sin \theta \\ dx dy &= r dr d\theta \end{aligned} \quad \int_0^{2\pi} \int_0^1 (1+r^2)^{3/2} r dr d\theta = \int_0^{2\pi} d\theta \cdot \int_0^1 (1+r^2)^{3/2} r dr$$

$$\begin{aligned} \int_0^1 (1+r^2)^{3/2} r dr &= \frac{1}{2} \int_0^1 (1+r^2)^{3/2} 2r dr = \frac{1}{2} \left[\frac{2}{5} (1+r^2)^{5/2} \right]_0^1 = \frac{4\sqrt{2}}{5} - \frac{1}{5} = \frac{4\sqrt{2}-1}{5} \\ u &= 1+r^2 \\ du &= 2r dr \end{aligned}$$

$$\int_0^{2\pi} d\theta = 2\pi \rightarrow \int_0^{2\pi} \int_0^1 (1+x^2+y^2)^{3/2} dx dy = 2\pi \left(\frac{4\sqrt{2}-1}{5} \right) = \frac{2\pi}{5} (4\sqrt{2}-1)$$

18. Evaluar $\int_S \frac{dx dy dz}{(x^2 + y^2 + z^2)^{3/2}}$, donde S es el sólido acotado por las dos esferas

$$x^2 + y^2 + z^2 = a^2 \text{ y } x^2 + y^2 + z^2 = b^2, \text{ donde } 0 < b < a.$$

$$\int_S (x^2 + y^2 + z^2)^{-3/2} dx dy dz$$

Aplico coordenadas esféricas:

$$x = \rho \sin(\phi) \cos(\theta)$$

$$y = \rho \sin(\phi) \sin(\theta)$$

$$z = \rho \cos(\phi)$$

$$\rho \in [b, a]$$

$$\theta \in [0, 2\pi]$$

$$\phi \in [0, \pi]$$

$$x^2 + y^2 + z^2 = \rho^2$$

$$dx dy dz = \rho^2 \sin(\phi) d\rho d\theta d\phi$$

$$\int_0^\pi \int_0^{2\pi} \int_b^a \rho^{-3/2} \rho^2 \sin(\phi) d\rho d\theta d\phi = \int_0^\pi \int_0^{2\pi} \int_b^a \rho^{-1/2} \sin(\phi) d\rho d\theta d\phi$$

$$= \int_0^\pi \sin(\phi) d\phi \cdot \int_0^{2\pi} d\theta \cdot \int_b^a \rho^{-1/2} d\rho = [-\cos(\phi)]_0^\pi \cdot [\theta]_0^{2\pi} \cdot [2\rho^{1/2}]_b^a$$

$$= 2 \cdot 2\pi \cdot \ln\left(\frac{a}{b}\right) = 4\pi \ln\left(\frac{a}{b}\right)$$

19 Evaluar lo siguiente usando coordenadas cilíndricas.

(a) $\iiint_B z \, dx \, dy \, dz$ donde B es la región dentro del cilindro $x^2 + y^2 = 1$, sobre el plano xy y debajo del cono $z = (x^2 + y^2)^{1/2}$

(b) $\iiint_D (x^2 + y^2 + z^2)^{-1/2} \, dx \, dy \, dz$ donde D es la región determinada por las condiciones $\frac{1}{2} \leq z \leq 1$ y $x^2 + y^2 + z^2 \leq 1$

$$a) \iiint_B z \, dx \, dy \, dz$$

Coordenadas cilíndricas:

$$x = r \cos(\theta)$$

$$y = r \sin(\theta)$$

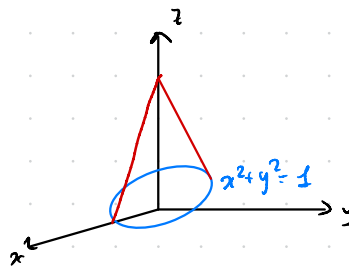
$$z = z$$

$$dx \, dy \, dz = r \, dr \, d\theta \, dz$$

$$r \in [0, 1]$$

$$\theta \in [0, 2\pi]$$

$$z \in [0, r]$$



$$\int_0^1 \int_0^{2\pi} \int_0^r z \, dz \, d\theta \, dr = \int_0^1 \int_0^{2\pi} \left[\frac{rz^2}{2} \right]_0^r dr \, d\theta = \int_0^1 \int_0^{2\pi} \frac{r^3}{2} dr \, d\theta$$

$$= \int_0^1 \frac{r^3}{2} dr \cdot \int_0^{2\pi} d\theta = \left[\frac{r^4}{8} \right]_0^1 \cdot [\theta]_0^{2\pi} = \frac{1}{8} \cdot 2\pi = \frac{\pi}{4}$$

$$b) \iiint_D (x^2 + y^2 + z^2)^{-1/2} = \int_{1/2}^1 \int_0^{2\pi} \int_0^{\sqrt{1-z^2}} \frac{r}{\sqrt{r^2+z^2}} dr = \int_{1/2}^1 \int_0^{2\pi} \left[\frac{1}{2} \sqrt{r^2+z^2} \right]_0^{\sqrt{1-z^2}} d\theta \, dz$$

$$x = r \cos \theta \quad r \in [0, \sqrt{1-z^2}]$$

$$y = r \sin \theta \quad \theta \in [0, 2\pi]$$

$$z = z \quad z \in [1/2, 1]$$

$$dx \, dy \, dz = r \, dr \, d\theta \, dz$$

$$= \int_{1/2}^1 \int_0^{2\pi} \frac{1}{2} \left[\sqrt{1-z^2} - \frac{z}{2} \right] d\theta \, dz = \frac{1}{2} \int_{1/2}^1 \left[\theta - \frac{z}{2} \theta \right] dz$$

$$= \frac{1}{2} \int_{1/2}^1 2\pi - \pi z \, dz = \frac{\pi}{2} \left[2z - \frac{z^2}{2} \right]_{1/2}^1 = \frac{5\pi}{16}$$

19. Usando coordenadas esféricas calcular la integral de $f(\rho, \phi, \theta) = 1/\rho$ sobre la región en el primer octante de $\mathbb{R}^3$ que está acotada por los conos $\phi = \pi/4$, $\phi = \arctan 2$ y la esfera $\rho = \sqrt{6}$.

Coordenadas esféricas:

$$x = \rho \cos(\theta) \sin(\phi)$$

$$y = \rho \sin(\theta) \sin(\phi)$$

$$z = \rho \cos(\phi)$$

$$dx dy dz = \rho^2 \sin^2 \phi \, d\rho \, d\theta \, d\phi$$

$$\int_{\pi/4}^{\arctan 2} \int_0^{2\pi} \int_0^{\sqrt{6}} \frac{1}{\rho} \rho^2 \sin^2(\phi) \, d\rho \, d\theta \, d\phi = \int_0^{2\pi} \int_{\pi/4}^{\arctan 2} \int_0^{\sqrt{6}} \rho \sin^2 \phi \, d\rho \, d\phi \, d\theta$$

$$= \int_0^{2\pi} \int_{\pi/4}^{\arctan 2} \left[\frac{\rho^2}{2} \sin^2 \phi \right]_0^{\sqrt{6}} d\phi \, d\theta = \int_0^{2\pi} \int_0^{\arctan 2} 3 \sin^2 \phi \, d\phi \, d\theta = \int_0^{2\pi} d\theta \cdot \int_{\pi/4}^{\arctan 2} 3 \sin^2 \phi \, d\phi$$

$$\int_{\pi/4}^{\arctan 2} 3 \sin^2 \phi \, d\phi = \frac{3}{2} \int_{\pi/4}^{\arctan 2} 1 - \cos 2\phi \, d\phi = \frac{3}{2} \left[\phi - \frac{1}{2} \sin 2\phi \right]_{\pi/4}^{\arctan 2}$$

$$= \frac{3}{2} \arctan 2 - \frac{3}{5} - \frac{3\pi}{8} + \frac{3}{4} = \frac{3}{2} \arctan 2 + \frac{3}{20} - \frac{3\pi}{8}$$